



Effects of prenatal exposure to air particulate matter on the risk of preterm birth and roles of maternal and cord blood LINE-1 methylation: A birth cohort study in Guangzhou, China

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ABSTRACT

Background: Epidemiological studies have found that increased risk of preterm birth (PTB) is associated with higher prenatal exposure to PM₁₀ and PM_{2.5}, but few studies have been conducted to assess the impacts of extremely fine particulate matter (PM₁) which may have more toxic effects than other types of ambient particulate air pollution (PM). Several studies have separately investigated the associations between DNA methylation and PTB risk and PM. Maternal LINE-1 methylation level negatively correlated with prenatal exposure to PM and risk of PTB. A comprehensive picture is lacking regarding the associations between prenatal exposure to PM, LINE-1 methylation, and risk of PTB.

Objectives: This study aimed to estimate the effects of exposure to ambient PM (PM₁₀, PM_{2.5}, and PM₁) of different sizes during pregnancy on risk of PTB, identify susceptible exposure windows, and illustrate the roles of LINE-1 methylation in the associations between PM and PTB risk.

Methods: The Birth Cohort Study on Prenatal Environments and Offspring Health (PEOH) has been ongoing since 2016 in Guangzhou, China. A total of 4928 pregnant women were recruited during early pregnancy, and 4278 (86.8%) were successfully followed-up. Each individual weekly exposure to PM₁₀ and PM_{2.5} from 3 months before pregnancy to childbirth was assessed using a spatiotemporal land use regression model, and the weekly PM₁ exposure was estimated by employing a generalized additive model. Maternal and cord blood LINE-1 methylation levels (%5mC) were tested using bisulfite-PCR pyrosequencing. A distributed lag nonlinear model incorporated with a Cox proportional hazard model was applied to assess the effect of weekly-specific maternal PM exposure on PTB risk, and a multiple-linear regression model was employed to investigate the associations between PM exposure and LINE-1 methylation levels of maternal and cord bloods. We also assessed the associations between LINE-1 methylation levels and PTB risk by using a logistic regression model.

Results: The risk of PTB was positively associated with PM_{2.5} and PM₁ concentrations during the 12th to 20th gestational weeks, and the strongest association was in the fourth quartile (Q₄) versus the first quartile (Q₁) and observed during the 16th gestational week (PM_{2.5}: hazard ratio [HR] = 1.18, 95%CI: 1.04–1.35, IQR = 11.94 µg/m³. PM₁: HR = 1.20, 95%CI: 1.03–1.39, IQR = 11.36 µg/m³). We observed significantly negative associations of PM₁₀ (β = −0.51%5mC per 10 µg/m³, P = 0.014), PM_{2.5} (β = −0.66%5mC per 10 µg/m³, P = 0.032) and PM₁ (β = −0.67%5mC per 10 µg/m³, P = 0.032) concentrations with cord blood LINE-1

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methylation levels, and a negative association between PM₁ concentration and maternal LINE-1 methylation level ($\beta = -0.86\%5\text{mC per } 10 \mu\text{g}/\text{m}^3$, $P = 0.034$).

Conclusion: Higher prenatal exposure to PM₁ and PM_{2.5} during the 12th to 20th gestational weeks was associated with increased risk of PTB. Maternal and fetal LINE-1 methylation alternation might be an underlying mechanism of PM that increasing the risk of PTB.

1. Introduction

Preterm birth (PTB) is all births before 37 weeks of pregnancy (WHO, 2018). Complications of PTB are the leading cause of mortality in children aged under 5 years (WHO, 2018). Survived preterm neonates are also at higher risk of many sequelae from infancy to adulthood, including neurodevelopmental sequelae, behavioral sequelae, and chronic health disorders (WHO, 2012; Saroj and Doyle, 2008). The global rate of PTB in 2014 was estimated at 10.6% (Chawanpaiboon et al., 2019). In China, the rate of PTB was 6.9% in 2014, accounting for 7.8% of PTB globally (Chawanpaiboon et al., 2019).

The causes of PTB are extremely complex. Studies have demonstrated that PTB was associated with maternal disease history, life behaviors, placental factors, psychological stress, and genetic factors (Goldenberg et al., 2008; Rich-Edwards and Grizzard, 2005; Wadhwa et al., 2010). In recent years, many epidemiological studies have observed the associations between air pollution and the risk of PTB (Sapkota et al., 2012; Zhang and Jin, 2008; Sun et al., 2015). Fetuses are in a critical period of growth and development, rapid cell proliferation, and immature organ development and metabolic changes, which makes them susceptible to toxic substances in the environment (Backes et al., 2013; Guo and Wei, 2013). For instance, our meta-analysis included 18 studies that had observed that each $10 \mu\text{g}/\text{m}^3$ increase in prenatal exposure to ambient PM_{2.5} was associated with a 13% increment in the risk of PTB (Sun et al., 2015). However, most studies investigated the effects of PM₁₀ and PM_{2.5}. Few studies have assessed the impacts of extremely fine particulate matter (PM₁), which may have more toxic effects than other types (Gong et al., 2014).

Identification of susceptible exposure windows will facilitate improvement of prenatal care and effective implementation of intervention by clinicians. Although many studies have attempted to identify the susceptible windows, the findings have been controversial. For example, Pereira et al. (2014a) observed that PM_{2.5} exposure during the first and the third trimesters were significantly associated with PTB. DeFranco et al. (2016) and Cheng et al. (2016) found that the third trimester might be the susceptible window. Guo et al. (2018) reported that PTB was significantly associated with PM_{2.5} exposure in any trimester. Another study suggested that trimester-specific association studies may lead to biased findings and incorrect estimation of susceptible windows, because biological changes in air pollution may not totally follow trimester periods (Wilson et al., 2017). A novel statistical method used a distributed lag model (DLM) and was proposed to estimate the associations between air pollution and pregnancy outcomes based on weekly exposure data (Wilson et al., 2017). In one of our studies, a retrospective study, we employed this method to assess the association between weekly air pollution exposure and PTB risk (Wang et al., 2018a, 2018b).

Biological mechanisms of air pollution leading to PTB are not completely understood, although studies have demonstrated that the mechanisms may include oxidative stress, inflammatory and hemodynamic responses, and obstruction of normal placental development (Vadillo-Ortega et al., 2014; Pereira et al., 2014b). Several recent studies have suggested that epigenetic changes, especially DNA methylation, which was the most intensely studied, may be one mechanism by which air pollution exerts its effects (Burris et al., 2012; McDonald et al., 2005). DNA methylation could affect gene expression and, thus, change biological function without modifying the DNA sequence.

Several previous studies have investigated the role of DNA

methylation of long interspersed nucleotide element (LINE)-1 in the association between PTB risk and exposure to particulate air pollution (PM). LINE-1 is an evolutionarily conserved superfamily, mainly distributed in the A(adenine)-T(thymine) enrichment region. The decrease in promoter methylation level can induce the activation and transcription of retrotransposons, which may trigger the immune system and induce inflammatory response (Crow, 2010). Inflammatory response is an important initiating mechanism of PTB (Vadillo-Ortega et al., 2014). Epidemiological studies have observed that PM_{2.5} exposure during pregnancy negatively correlated with placental global DNA methylation level (Janssen et al., 2013), and that LINE-1 methylation decreased after exposure to PM_{2.5} in the elderly individuals (Baccarelli et al., 2009). Other studies have found that maternal LINE-1 methylation level was negatively correlated with PTB (Burris et al., 2012). These results suggest that LINE-1 methylation alternation may be an important biological mechanism underlying the effects of PM on PTB risk. However, no epidemiological evidence has been provided on direct relationships between PM exposure, LINE-1 methylation, and PTB.

In this study, we conducted the Birth Cohort Study on Prenatal Environments and Offspring Health (PEOH) in Guangzhou, China in 2016. We aimed to prospectively estimate the effects of prenatal exposure to PM of different sizes (*i.e.*, PM₁₀, PM_{2.5}, and PM₁) on the risk of PTB, identify the susceptible exposure windows, and further illustrate the roles of both maternal and umbilical cord blood LINE-1 methylation in the associations between PM and risk of PTB.

2. Material and methods

2.1. Study setting and participants

The PEOH was conducted to investigate associations between maternal exposure to environmental factors and adverse fetal health and birth outcomes and further illustrate the underlying epigenetic mechanisms. Guangzhou Panyu Central Hospital was the study setting, the largest hospital in Panyu district, Guangzhou. This birth cohort study was initiated in January 2016. All pregnant women were initially recruited from the outpatient department of obstetrics if they fulfill the following criteria: (1) gestational weeks from 1 to 13; (2) aged 18 to 50 years; (3) and the following diseases were not present: hyperthyroidism, heart disease, chronic kidney disease, tuberculosis, psychiatric disease and other serious diseases. The PEOH was approved by the Ethics Committee of Guangdong Provincial Center for Disease Control and Prevention. Every recruited participant was provided a detailed introduction and explanation of this study, and signed the informed consent.

2.2. Baseline investigation

Each participant engaged in a face-to-face interview administered by a trained public health professional and based on a questionnaire. The main contents of the questionnaire were maternal demographic characteristics, life behaviors, family addresses and changes, household living circumstances, work environments, activity patterns, history of diseases, and diets. The participants' prenatal care records were extracted from the hospital information system. All data were saved as the baseline information, and the follow-up profile was established for each participant.

2.3. First follow-up investigation

The first follow-up investigation was conducted for each participant who had to finish the second questionnaire during her hospital delivery. Maternal blood samples and cord blood samples were collected. All collected blood samples were immediately refrigerated and transferred to the laboratory for storage at -80°C for future laboratory tests. Birth records of each neonate were extracted from the maternal medical record. This follow-up investigation was completed before December 2017.

2.4. Endpoints

The gestational age in weeks was estimated by the ultrasound measurement, or calculated by subtracting the last menstrual period (LMP) date from the birth date. When the results between the LMP based estimation and ultrasound data were inconsistent, the ultrasound examination would prevail. All infants were defined as PTBs if their gestational ages were < 37 completed weeks.

A total of 4928 pregnant women were recruited in the baseline investigation, and 4278 (86.8%) were successfully followed-up during the hospitalization for childbirth. We excluded multiple births ($N = 78$), stillbirths ($N = 8$), and participants with missing information for key variables such as gestational age ($N = 24$), infant sex ($N = 6$), maternal age ($N = 3$), maternal education ($N = 3$), and maternal prepregnancy body mass index (BMI, $N = 7$). Finally, 4101 participants (234 preterm cases) were included in this analysis to estimate the associations between maternal exposure to PM and the risk of PTB, and identify susceptible exposure windows.

To further illustrate the roles of LINE-1 methylation in the associations between PM and PTB risk, we conducted two nested case-control studies. In the first study, we first selected all the participants ($N = 2129$) whose blood samples were collected at childbirth. Among these participants, all preterm cases ($N = 70$) were selected as cases, and controls ($N = 119$) were randomly selected from the term births. Similarly, in the second nested case-control study, all participants whose cord blood samples were collected were selected ($N = 2923$). Next, all preterm cases ($N = 113$) were selected as a case group, and the control group ($N = 145$) was randomly selected from the remainder of the participants.

2.5. Air pollution exposure assessment

First, we collected daily ambient air pollutant data (PM_{10} , $\text{PM}_{2.5}$, O_3 , SO_2 , NO_2 , and CO) of 102 monitoring stations in Guangdong province from the National Urban Air Quality Real-time Publishing Platform (<http://106.37.208.233:20035/>) (January 1, 2014, to December 31, 2017), meteorological data (daily mean temperature [TM], relative humidity [RH], mean wind speed [WS], atmospheric pressure [AP], and visibility) of all 86 monitoring stations from Guangdong Meteorological Service, population density data in 2015 from the GeoData Institute at the University of Southampton (www.worldpop.org.uk), and geographic information system covariates (geographic map, road density, and land use data) from the Data Center for Resources and Environmental Sciences (<http://www.resdc.cn>). Detailed information on the data is in the supplementary material (Table S1).

Second, we prepared the data for the spatiotemporal land use regression (LUR) model. We described the detailed process of $\text{PM}_{2.5}$ exposure assessment in a previous study (Liu et al., 2019a, 2019b). We extracted the latitude and longitude information for each subject's residential and work place address and air quality monitoring station. An inverse distance weighted method was used to create a surface and estimate the daily meteorological data (TM, RH, WS, AP and visibility) at a 1×1 km spatial resolution across the entire province. Next, we extracted the daily meteorological data from the grid where the

participant's residence and workplace addresses overlapped, and calculated the weekly average meteorological data during the entire pregnancy period of each participant. We further extracted the population density, length of road, and types of land use data for each address by using a radius of 1300 m, selected based on a series of pilot examinations. The results of the pilot assessments of $\text{PM}_{2.5}$ exposure were described in detail in our previous study (Liu et al., 2019a, 2019b).

Third, we established a spatiotemporal LUR model using the prepared data of all air quality monitoring stations: weekly air pollutants, weekly visibility, population density, road length, and land use data. Two smooth temporal basis functions were added to the model. In particular, particulate matter was demonstrated to scatter or absorb light and, hence, reduces visibility (Ge et al., 2011). Many studies have suggested that visibility was closely correlated with air quality and could be used as an alternative indicator of air quality in health research (Ge et al., 2011; Liu et al., 2014; Tie et al., 2009). Therefore, visibility was an important covariate in the LUR model. The process of model establishment has been well demonstrated in an R package *SpatioTemporal*. The result of 10-fold cross-validation for the spatiotemporal LUR model on $\text{PM}_{2.5}$ exposure assessment showed that the R^2 was 88.86% and the root mean square error (RMSE) was $5.53 \mu\text{g}/\text{m}^3$, indicating good performance of model fitness. Fourth, we input the parameters of each participant's address into the model and predicted the weekly $\text{PM}_{2.5}$ from January 1, 2014, to December 31, 2017. Similarly, we employed the spatiotemporal LUR model to estimate the weekly average concentrations of PM_{10} , O_3 , SO_2 , NO_2 , and CO . The performance of the LUR modeling is shown in Fig. S1.

Because ambient PM_{10} has not been widely monitored in China, based on the high correlation between PM_{10} and $\text{PM}_{2.5}$ concentrations and their relationships with other meteorological data, we employed a generalized additive model (GAM) to predict the weekly PM_{10} concentrations of each participant's address by using the predicted weekly $\text{PM}_{2.5}$, TM, RH, AP, and WS. This GAM model was described in one of our previous studies, that is, the R^2 for daily prediction was 93% (Chen et al., 2017).

$$\text{PM}_{10} = s(\text{PM}_{2.5}) + s(\text{TM}) + s(\text{RH}) + s(\text{AP}) + s(\text{WS}) \quad (1)$$

where, TM, RH, AP, and WS refer to daily mean temperature, relative humidity, mean atmospheric pressure, and wind speed. The degrees of freedom for smooth terms were automatically selected by GAM.

Based on these estimated $\text{PM}_{2.5}$ concentrations, we used Eq. (2) to estimate the time-weighted individual exposure to $\text{PM}_{2.5}$ during workdays and employed Eq. (3) to assess the exposure during non-workdays based on each participant's indoor and outdoor activity time (Chen et al., 2013). For each individual, the weekly exposure (from the first day of the corresponding gestational week to the following 6 days) from 12 weeks before the LMP date to the week of delivery was calculated.

$$\begin{aligned} Y_{\text{air}} &= (C_{\text{hout}} * T_{\text{hout}} + C_{\text{wout}} * T_{\text{wout}} + IF_h * (C_{\text{hout}} * T_{\text{hin}} + C_{\text{wout}} * T_{\text{win}}) \\ &\quad + T_{\text{trans}} * IF_t * (C_{\text{hout}} + C_{\text{wout}}))/2/24 \end{aligned} \quad (2)$$

$$Y_{\text{air}} = (C_{\text{hout}} * T_{\text{hout}} + IF_h * C_{\text{hout}} * T_{\text{hin}} + T_{\text{trans}} * IF_t * C_{\text{hout}})/24 \quad (3)$$

where Y_{air} denotes the time-weighted $\text{PM}_{2.5}$ concentration of each individual; C_{hout} denotes the average concentration of ambient $\text{PM}_{2.5}$ at each participant's residence address, which was already estimated; C_{wout} denotes the average concentration of ambient $\text{PM}_{2.5}$ at each participant's workplace which was already estimated; T_{hout} denotes the daily average hours spent outdoors around their residence; T_{wout} denotes the daily average hours spent outdoors around their workplace; T_{hin} denotes the daily average hours spent indoors at their residence; T_{win} denotes the daily average hours spent indoors in their workplace; T_{trans} denotes the daily average hours spent on transportation; IF_h denotes the

infiltration factor (0.83) of $PM_{2.5}$ infiltrating into indoor environments from the outdoors (Zhou et al., 2018); IF_t denotes the infiltration factor of $PM_{2.5}$, which varies by type of transportation (Table S2). Here, we employed the average concentration of $PM_{2.5}$ at the residence and workplace to represent the $PM_{2.5}$ concentration for transportation. We used the same method to assess the exposure to other air pollutants for each participant. The entire process of air pollution exposure assessment is presented in Fig. S2.

This study was approved by the Review Board of Guangdong Provincial Center for Disease Control and Prevention. The data used in this study were anonymous; thus, identifiable personal information was not used. The PEOH was registered in the Chinese Clinical Trial Registry (ChiCTR-ROC-17013496).

2.6. Covariates

The following variables were considered potential confounders: maternal socio-demographic characteristics (maternal age, education, marital status, occupation, and household income), lifestyle factors (maternal smoking, passive smoking, paternal smoking, tea consumption and alcohol consumption during pregnancy), infant sex, and maternal medical information (prepregnancy BMI, parity, gravidity, season of conception, gestational hypertension, gestational diabetes, and adverse pregnancy history). The definition of these variables is shown in Table S3. Ambient air temperature and other air pollutants were also considered as potential confounders.

2.7. LINE-1 methylation measurement

Two nested case-control studies were used to estimate the associations between LINE-1 methylation in maternal blood or cord blood with PM exposure and the risk of PTB. Therefore, only the PTB case and control samples were selected for LINE-1 methylation measurement. The DNA of maternal and umbilical cord blood were extracted using the RelaxGene Blood DNA System (0.1–20 mL) (DP319) (Tiangen Biotech Beijing CO Ltd). Notably, 2 μ g genome DNA was used for bisulfite conversion by employing the EpiTect bisulfite Kit (Qiagen), and the temperature program of bisulfite conversion was exhibited in Table S4. The DNA methylation levels were quantitated by pyrosequencing, which has been described in the literature (Duan et al., 2013; Bollati et al., 2007). In brief, 50 ng bisulfite modified DNA was amplified by pyrosequencing PCR (Duan et al., 2013; Pavanello et al., 2010) (see Table S4 for PCR primers and conditions). Additionally, 5 μ L amplified PCR product was electrophoresed to identify the specificity of the PCR product. Sequencing was performed only if the PCR product had no nonspecific bands and no primer dimers. The pyrosequencing was performed using the PyroMark Q96 ID system (Qiagen) according to the protocol of the manufacturer. We used the percentage of methylated cytosines divided by the sum of methylated and nonmethylated cytosines to express the degree of DNA methylation. All samples were tested twice, and their average methylation levels were used in the statistical analysis (Baccarelli et al., 2009). Furthermore, sentinel quality control was set for each batch of sequencing. If the variability of sentinel samples tested by different batches was < 10%, the sequencing results were considered reliable and could be used for subsequent statistical analysis.

2.8. Statistical analysis

First, a Chi-square test and *t*-test were used to assess the differences in characteristics between the case group and control group. We used Pearson's correlation to estimate the correlations among PM_{10} , $PM_{2.5}$, PM_1 , SO_2 , CO, O_3 , and NO_2 in all participants.

Second, a distributed lag nonlinear Model (DLNM) incorporated with a Cox proportional hazard model was applied to assess the exposure-lag-response between weekly-specific maternal PM exposure as

continuous variables or categorical variables divided by quartiles and risk of PTB by accounting for the lagged exposure in all included participants. We first employed a time-by-covariate interactions method to evaluate the assumptions of the Cox proportional hazards model for the association between the time-dependent covariate PM (i.e., PM_{10} , $PM_{2.5}$ and PM_1) and PTB risk (David, 1972). The results showed that the PH (proportional hazard) assumption was fulfilled for all PM variables, and the *P* value of PM_{10} , $PM_{2.5}$, and PM_1 were 0.504, 0.473 and 0.572, respectively. This method was described in our previous study (Wang et al., 2018a, 2018b).

By defining both traditional exposure-response and the additional lag-response association, the model can describe complex exposure-lag-response by creating a cross-basis function (Gasparrini, 2014). In this study, a liner function was used to model the exposure-response relationship and a natural cubic spline function was used to model the lag-response relationship between PTB and PM as continuous and categorical variables, respectively. We applied a Cox PH model to respectively estimate the HR and 95% confidence intervals (95%CI) for each 10 μ g/ m^3 increase in exposure to PM concentrations or changes from the first quartile (reference) to the other three quartiles. We assumed the associations between PM exposure and PTB were smooth across weeks for the lag-response associations (Han et al., 2018; Zhang et al., 2012). A natural cubic spline (knots at equally-spaced values of lag range) was selected to model the lag-response relationship. The optimal *df* of lag structure was selected based on minimum Akaike Information Criterion (Gasparrini, 2011, 2014). We set 48 weeks as the maximum lag period, and confined -12~ week 0 (before LMP) and 1–37 weeks (gestational weeks) as exposure periods (Han et al., 2018). Finally, the *dfs* used for lag distribution was 5 for PM_{10} , $PM_{2.5}$ and PM_1 . Weekly mean ambient temperature during pregnancy was adjusted by using a natural cubic spline (*df* = 3).

Third, we investigated the associations between PM exposure and LINE-1 methylation levels of maternal or cord bloods by using multiple-linear regression models in the two nested case-control studies. The PM exposure was the average level of PM concentration during the periods where was significantly associated with the risk of PTB. The exposure to PM was also treated as continuous variables or categorical variables divided by quartiles. In the multiple-linear regression model, we adjusted for the variables whose distributions differed significantly between the cases and controls in the total participants as potential confounding factors. These variables included maternal age and parity. However, several studies have suggested that other variables must be controlled for the associations between PM and LINE-1 methylation, including gestational weeks, paternal smoking, prepregnancy BMI, gestational hypertension, gestational diabetes, adverse pregnancy history, and infant sex (Baccarelli et al., 2009; Janssen et al., 2013). Therefore, we also added these covariates in the list of adjusted for variables. We also assessed the associations between LINE-1 methylation level as a continuous variable or a categorical variable with PTB risk by using a logistic regression model adjusted for potential confounders.

2.9. Sensitivity analysis

Sensitivity analyses were performed to test the robustness of the results. We changed the *df* of lag distribution for PM exposure from 4 to 6 in the DLNM model to test the robustness of PM effects. In addition, we employed a two-pollutant model to include other air pollutants (CO , O_3 or SO_2) one by one to evaluate their influences on the effects of PM. We did not include NO_2 due to its high correlation with PM, which may lead to potential collinearity (Table S6). PM_{10} , $PM_{2.5}$ and PM_1 were not concurrently included in one model because of their close correlations.

In addition, we employed different buffer radiuses to extract the population density, length of road, and types of land use data, and test the impacts of radius on the performance of the spatiotemporal LUR model. Based on our previous study (Liu et al., 2019a, 2019b), we employed 1000, 1300, and 1500 m as the radiuses of population

density, length of road, and types of land use data, respectively.

All statistical analyses were performed using R3.4.0 (R Development Core Team 2017, <https://www.r-project.org>), and packages “dlnm,” “splines,” “survival,” “foreign,” “car,” “lubridate,” and “psych” in R software were used (Gasparrini, 2014). The statistical tests were two-sided, and a *p* value of < 0.05 was considered statistical significance.

3. Results

3.1. General characteristics of participants

A total of 4101 participants were included in the final analyses, with an average age of 32.4 years. The incidence rate of PTB was 5.7% (234/4101). Compared with the term births, PTBs occurred more frequently in women with an older age, more parities, higher prepregnancy BMI, more exposure to passive smoking, greater proportion of gestational hypertension, and less tea consumption (Tables 1, 2). The weekly average concentrations of PM₁₀, PM_{2.5}, and PM₁ during the entire study

period were 48.67 µg/m³, 32.07 µg/m³ and 27.93 µg/m³, respectively. The weekly mean exposure to temperature was 23.46 °C (Table 3). Weekly mean concentrations of PM₁₀, PM_{2.5}, and PM₁ were significantly correlated with weekly mean temperature (Table S12).

Compared with the total participants, pregnant women selected in the nested case-control studies were older, and had higher parity and a larger proportion of gestational hypertension (Tables 1, 2). In the nested case-control studies, the proportion of participants with higher maternal blood LINE-1 methylation was lower in preterm cases (47.1%) than in controls (51.3%), but the proportion of participants with higher cord blood LINE-1 methylation was higher in preterm cases (54.9%) than in controls (46.2%) (Tables 1, 2).

3.2. Associations between PM exposure and PTB in total participants

The single-pollutant analyses in Fig. 1 show that, although no significant association was observed between weekly PM₁₀ exposure and PTB, the risk of PTB was positively associated with PM_{2.5} and PM₁

Table 1

Characteristics of preterm births and term births in the total participants and nested case-control studies.

	Total participants				Nested case-control study I ^a				Nested case-control study II ^b			
	Preterm births N (%)	Term births N (%)	χ ²	P	PTB cases N (%)	Controls N (%)	χ ²	P	PTB cases N (%)	Controls N (%)	χ ²	P
Maternal age (years)												
≤ 25	6(2.6)	209(5.4)	15.41	0.001	3(4.3)	3(2.5)	–	0.010	5(4.4)	3(2.1)	–	0.324
25–30	65(27.8)	1300(33.6)			20(28.6)	29(24.4)			31(27.4)	40(27.6)		
31–35	78(33.3)	1365(35.3)			18(25.7)	58(48.7)			38(33.6)	62(42.8)		
> 35	85(36.3)	993(25.7)			29(41.4)	29(24.4)			39(34.5)	40(27.6)		
Infant sex												
Male	142(60.7)	2030(52.5)	5.61	0.018	36(51.4)	71(59.7)	0.90	0.341	72(63.7)	76(52.4)	2.87	0.090
Female	92(39.3)	1837(47.5)			34(48.6)	48(40.3)			41(36.3)	69(47.6)		
Parity												
1	51(21.8)	1191(30.8)	37.04	< 0.001	13(18.6)	11(9.2)	6.48	0.039	28(24.8)	22(15.2)	10.66	0.005
2	133(56.8)	2310(59.8)			45(64.3)	71(59.7)			69(61.1)	79(54.5)		
> 2	50(21.4)	366(9.4)			12(17.1)	37(31.1)			16(14.1)	44(30.3)		
Gravidity												
1	66(28.2)	1144(29.6)	1.72	0.632	22(31.4)	37(31.1)	5.33	0.149	36(31.9)	43(29.7)	4.34	0.227
2	98(41.9)	1476(38.2)			32(45.7)	44(37.0)			44(38.9)	45(31.0)		
3	44(18.8)	832(21.5)			7(10.0)	27(22.7)			22(19.5)	31(21.4)		
> 3	26(11.1)	415(10.7)			9(12.9)	11(9.2)			11(27.7)	26(17.9)		
Marital status												
Married	225(96.2)	3766(97.4)	–	0.292	69(98.6)	111(93.3)	–	0.158	109(96.5)	141(97.2)	–	0.733
Others	9(3.8)	101(2.6)			1(1.4)	8(6.7)			4(3.5)	4(2.8)		
Maternal occupation												
Manual worker	13(5.6)	199(5.2)	–	0.968	3(4.3)	11(9.2)	–	0.202	5(4.4)	9(6.2)	–	0.155
Government official and clerk	5(2.1)	78(2.0)			2(2.9)	5(4.2)			2(1.8)	8(5.5)		
Housewife	20(8.5)	380(9.8)			7(10.0)	11(9.2)			9(8.0)	13(9.0)		
Unemployment	17(7.3)	284(7.3)			5(7.1)	11(9.2)			9(8.0)	8(5.5)		
Technician	39(16.7)	724(18.7)			16(22.7)	15(12.6)			24(21.2)	19(13.1)		
Business	132(56.4)	2076(53.7)			36(51.4)	56(47.1)			62(54.9)	78(53.8)		
Others	8(3.4)	126(3.3)			1(1.4)	10(8.4)			2(1.8)	10(6.7)		
Household income (×1000Yuan)												
< 30	24(10.1)	188(4.9)	–	0.018	5(7.1)	13(10.9)	–	0.373	12(10.6)	16(11.0)	–	0.993
30~	138(59.0)	2315(59.9)			40(57.1)	66(55.5)			62(54.9)	80(55.2)		
100~	60(25.6)	1134(29.3)			23(32.9)	29(24.4)			33(29.2)	42(29.0)		
≥ 200	10(4.3)	193(5.0)			2(2.9)	8(6.7)			5(4.4)	5(3.5)		
Refused to answer/missing	2(0.9)	37(1.0)			0(0)	3(2.5)			1(0.9)	2(1.4)		
Maternal education (years)												
≤ 12	46(19.7)	603(15.6)	4.83	0.089	14(20.0)	22(18.5)	1.07	0.587	21(18.6)	19(13.1)	4.16	0.125
13–15	118(50.4)	2218(57.4)			30(42.9)	60(50.4)			52(46.0)	85(58.6)		
> 15	70(29.9)	1046(27.0)			26(37.1)	37(31.1)			40(35.4)	41(28.3)		
Season of conception												
Warm seasons (May to Oct.)	111(47.4)	1832(47.4)	< 0.01	1.000	36(51.4)	60(50.4)	0.02	1.000	61(54.0)	70(48.3)	0.61	0.433
Cold seasons (Nov. to Apr.)	123(52.6)	2035(52.6)			34(48.6)	59(49.6)			52(46.0)	75(51.7)		

– Fisher's exact test.

^a Cases and controls were selected from the participants whose maternal blood samples have been collected.

^b Cases and controls were selected from the participants whose cord blood samples have been collected.

Table 2
Characteristics of preterm cases and controls in the total participants and nested case-control studies (*continued*).

	Total study				Nested case-control study I ^a				Nested case-control study II ^b			
	Preterm births N (%)	Term births N (%)	χ^2	P	PTB Cases N (%)	Controls N (%)	χ^2/t	P	PTB Cases N (%)	Controls N (%)	χ^2/t	P
Prepregnancy BMI (kg/m ²)												
Under weight (< 18.5)	55(23.5)	989(25.6)	9.76	0.021	16(22.9)	26(21.8)	–	0.333	26(23.0)	26(17.9)	–	0.461
Normal weight (18.5–)	128(54.7)	2322(60.0)			42(60.0)	78(65.6)			64(56.6)	94(64.8)		
Over weight (24–)	42(18.0)	448(11.6)			12(17.1)	12(10.1)			17(15.0)	21(14.5)		
Obesity (≥ 28)	9(3.9)	108(2.8)			0(0)	3(2.5)			6(5.3)	4(2.8)		
Maternal smoking												
Yes	0(0)	4(0.1)	–	1.000	0(0)	0(0)	–	–	0(0)	0(0)	–	–
No	234(100.0)	3863(99.9)			70(100.0)	119(100.0)			113(100.0)	145(100.0)		
Passive smoking												
Yes	54(23.1)	1169(30.2)	5.06	0.025	12(17.1)	28(23.5)	0.73	0.393	33(29.2)	28(19.3)	2.92	0.088
No	180(76.9)	2698(69.8)			58(82.9)	91(76.5)			80(70.8)	117(80.7)		
Paternal smoking												
Yes	74(31.6)	1362(35.2)	1.10	0.294	19(27.1)	39(32.8)	0.42	0.518	34(30.1)	53(36.6)	0.92	0.339
No	160(68.4)	2505(64.8)			51(72.9)	80(67.2)			79(69.9)	92(63.4)		
Gestational hypertension												
Yes	24(10.3)	55(1.4)	–	< 0.001	7(10.0)	5(4.20)	–	0.131	8(7.1)	4(2.8)	–	0.137
No	210(89.7)	3812(98.6)			63(90.0)	114(95.90)			105(92.9)	141(97.2)		
Gestational diabetes												
Yes	67(28.6)	885(22.9)	3.77	0.050	20(28.6)	32(26.9)	0.007	0.935	31(27.4)	44(30.3)	0.14	0.709
No	167(71.4)	2982(77.1)			50(71.4)	87(73.1)			82(72.6)	101(69.7)		
Adverse pregnant history												
Yes	93(39.7)	1599(41.4)	0.17	0.677	19(27.1)	47(39.5)	2.44	0.118	40(35.4)	65(44.8)	1.97	0.161
No	141(60.3)	2268(58.6)			51(72.9)	72(60.5)			73(64.6)	80(55.2)		
Tea consumption												
Yes	11(4.7)	393(10.2)	–	0.005	1(1.4)	10(8.4)	–	0.056	7(6.2)	13(9.0)	0.35	0.554
No	223(95.3)	3474(89.8)			69(98.6)	109(91.6)			106(93.8)	132(91.0)		
Alcohol consumption												
Yes	1(0.4)	4(0.1)	–	0.255	0(0)	0(0)	–	–	0(0)	0(0)	–	–
No	233(99.6)	3863(99.9)			70(100.0)	119(100.0)			113(100.0)	145(100.0)		
LINE-1 methylation (%5mC)												
$\leq P_{50}^d$	–	–	–	–	37(52.9)	58(48.7)	0.16	0.692	51(45.1)	78(53.8)	1.57	0.210
$> P_{50}^d$	–	–	–	–	33(47.1)	61(51.3)			62(54.9)	67(46.2)		
LINE-1 methylation (%5mC, Mean (SD))	–	–	–	–	84.68(2.38)	84.57(3.04)	0.28 ^c	0.784	85.22(2.31)	84.85(2.87)	1.15 ^c	0.250

BMI, body mass index; SD, standard deviation.

–: Fisher's exact test.

^a Cases and controls were selected from the participants whose maternal blood samples have been collected.

^b Cases and controls were selected from the participants whose cord blood samples have been collected.

^c Statistics were estimated by *t*-test.

^d The median of LINE-1 methylation level was 84.72%5mC in the first nested case-control study, and 85.04%5mC in the second nested case-control study.

Table 3
Summary statistics of weekly exposure level to air pollutants and ambient temperature in the total participants during the entire study period.

	Mean	SD	Min	25th	50th	75th	Max	IQR
PM ₁₀ ($\mu\text{g}/\text{m}^3$)	48.67	14.61	11.67	38.57	45.88	56.19	135.06	17.62
PM _{2.5} ($\mu\text{g}/\text{m}^3$)	32.07	10.19	7.09	25.36	30.59	37.30	97.60	11.94
PM ₁ ($\mu\text{g}/\text{m}^3$)	27.93	9.46	4.97	21.63	26.71	32.99	85.75	11.36
SO ₂ ($\mu\text{g}/\text{m}^3$)	10.36	2.00	2.70	8.99	10.10	11.44	24.86	2.45
NO ₂ ($\mu\text{g}/\text{m}^3$)	36.38	10.45	5.93	28.50	34.22	42.05	95.00	13.55
CO($\mu\text{g}/\text{m}^3$)	0.72	0.11	0.24	0.65	0.70	0.78	1.38	0.13
O ₃ ($\mu\text{g}/\text{m}^3$)	44.40	14.61	6.36	34.37	43.92	54.31	99.24	19.93
Mean temperature (°C)	23.46	5.62	5.95	18.47	25.11	28.48	33.80	10.01

exposure during the 12th to 18th gestational weeks, which might be the susceptible exposure window, and the strongest association was observed during the 16th gestational week (PM_{2.5}: HR = 1.06 for per 10 $\mu\text{g}/\text{m}^3$, 95%CI: 1.01–1.11. PM₁: HR = 1.07 for per 10 $\mu\text{g}/\text{m}^3$, 95%CI: 1.01–1.13). Similar results were observed in the models in which the exposure to PM were divided by quartiles. Compared with the first quartile (Q₁) of PM_{2.5} and PM₁, a significantly higher risk of PTB was found for the fourth quartile (Q₄) during the 12th to 20th gestational

weeks, and the strongest associations were also observed during the 16th gestational week (PM_{2.5}: HR = 1.18, 95%CI: 1.04–1.35, IQR = 11.94 $\mu\text{g}/\text{m}^3$. PM₁: HR = 1.20, 95%CI: 1.03–1.39, IQR = 11.36 $\mu\text{g}/\text{m}^3$) (Fig. 1 and Table S13).

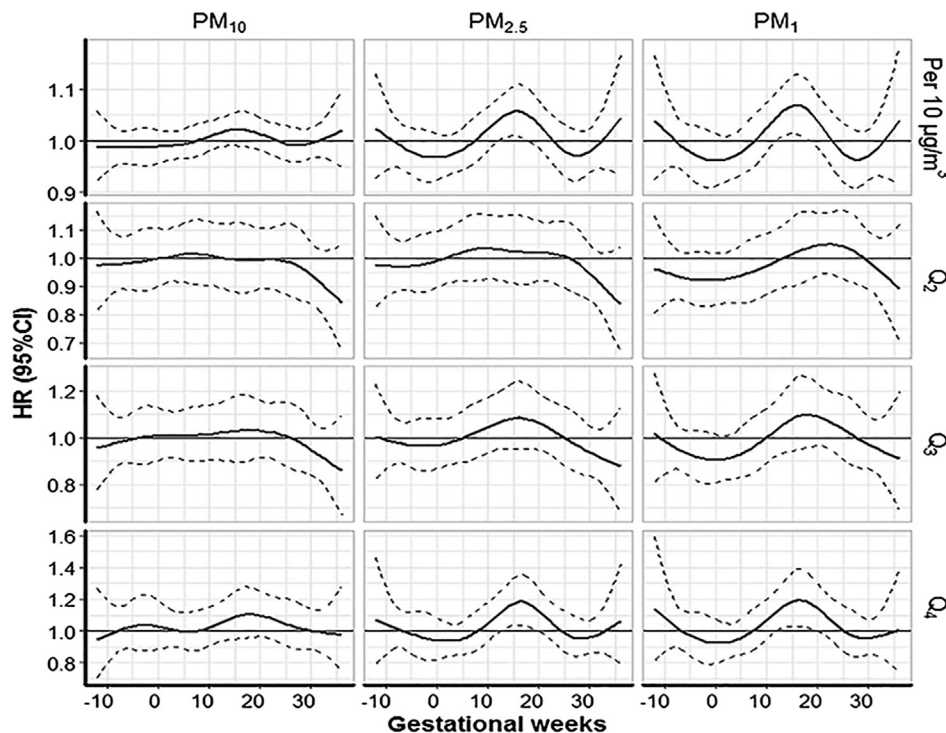


Fig. 1. Hazard ratio (HR) of PTB associated with weekly PM exposure from 12 weeks before LMP to the birth time.

HRs (95%CI) in the three uppermost charts indicate the risks of PTB for each $10 \mu\text{g}/\text{m}^3$ increment in weekly PM concentrations during pregnancy. HRs (95%CI) in the other charts indicate the risks of PTB for the second (Q2), third (Q3) and fourth quartiles (Q4) of weekly PM concentrations during pregnancy compared with the first quartile (Q1) as the reference.

Zero in the X-axis scale indicates the time of LMP (gestational week 0), minus numbers indicate the weeks prior to LMP, and plus numbers indicate the gestational weeks.

All models were adjusted for maternal age, gravidity, parity, infant sex, maternal occupation, household income, maternal education, marital status, season of conception, pre-pregnancy BMI, maternal smoking, passive smoking, paternal smoking, gestational hypertension, gestational diabetes, adverse pregnancy history, alcohol consumption, tea consumption, and weekly mean temperature.

3.3. Associations between PM exposure and LINE-1 methylation in the nested case-control studies

Maternal blood LINE-1 methylation level had significantly negative associations with average PM_{10} concentrations during the susceptible window. Every $10 \mu\text{g}/\text{m}^3$ increment in PM_{10} concentration was associated with an average of 0.86%5mC ($P = 0.034$) decrease in the LINE-1 methylation level. In addition, compared with Q1 of PM_{10} , the Q3 and Q4 decreased on average 1.21%5mC and 1.29%5mC in LINE-1 methylation, respectively (Table 4).

Cord blood LINE-1 methylation was significantly associated with PM_{10} , $\text{PM}_{2.5}$ and PM_1 exposure. Each $10 \mu\text{g}/\text{m}^3$ increment in PM_{10} , $\text{PM}_{2.5}$ and PM_1 concentrations was associated with an average decrease of 0.51 ($P = 0.014$), 0.66 ($P = 0.032$) and 0.67%5mC ($P = 0.032$) in the LINE-1 methylation levels, respectively. After dividing the PM concentrations into four groups by quartiles, we also observed a significantly lower LINE-1 methylation level in the Q4 group compared with the Q1 group of PM_{10} (Table 4).

Table 4

Associations (β , 95%CI) between PMs and LINE-1 methylation in the nested case-control studies.

	Nested case-control study I ^d					Nested case-control study II ^e			
	Crude β (95%CI)	P	Adjusted β (95%CI) ^a	P		Crude β (95%CI)	P	Adjusted β (95%CI) ^a	P
PM_{10}^b ($\mu\text{g}/\text{m}^3$)					PM_{10}^c ($\mu\text{g}/\text{m}^3$)				
Per $10 \mu\text{g}/\text{m}^3$	-0.43(-0.93,0.07)	0.095	-0.47(-0.98,0.04)	0.070	Per $10 \mu\text{g}/\text{m}^3$	-0.47(-0.86,-0.09)	0.016	-0.51(-0.91,-0.11)	0.014
Q1 (≤ 44.47)	Reference		Reference		Q1 (≤ 43.60)	Reference		Reference	
Q2 (44.47~)	-0.69(-1.82,0.45)	0.237	-0.51(-1.64,0.61)	0.372	Q2 (43.60~)	-0.43(-1.33,0.48)	0.357	-0.39(-1.33,0.54)	0.408
Q3 (47.84~)	-0.80(-1.92,0.33)	0.168	-0.87(-1.99,0.25)	0.130	Q3 (47.44~)	-0.21(-1.11,0.70)	0.652	-0.08(-1.03,0.87)	0.871
Q4 (≥ 56.70)	-1.03(-2.15,0.09)	0.073	-1.01(-2.14,0.13)	0.083	Q4 (≥ 56.59)	-1.08(-1.99,-0.18)	0.019	-1.13(-2.08,-0.19)	0.019
$\text{PM}_{2.5}^b$ ($\mu\text{g}/\text{m}^3$)					$\text{PM}_{2.5}^c$ ($\mu\text{g}/\text{m}^3$)				
Per $10 \mu\text{g}/\text{m}^3$	-0.74(-1.50,0.03)	0.06	-0.75(-1.53,0.02)	0.058	Per $10 \mu\text{g}/\text{m}^3$	-0.62(-1.19,-0.05)	0.033	-0.66(-1.25,-0.06)	0.032
Q1 (≤ 30.45)	Reference		Reference		Q1 (≤ 29.78)	Reference		Reference	
Q2 (30.45~)	0.04(-1.09,1.17)	0.946	-0.05(-1.19,1.10)	0.936	Q2 (29.78~)	-0.03(-0.93,0.88)	0.957	-0.07(-1.00,0.86)	0.888
Q3 (33.56~)	-0.55(-1.68,0.57)	0.338	-0.48(-1.64,0.67)	0.414	Q3 (32.85~)	-0.62(-1.53,0.28)	0.180	-0.64(-1.62,0.34)	0.205
Q4 (≥ 37.17)	-1.00(-2.13,0.12)	0.084	-0.96(-2.10,0.18)	0.099	Q4 (≥ 37.02)	-0.72(-1.63,0.18)	0.119	-0.78(-1.72,0.16)	0.106
PM_1^b ($\mu\text{g}/\text{m}^3$)					PM_1^c ($\mu\text{g}/\text{m}^3$)				
Per $10 \mu\text{g}/\text{m}^3$	-0.90(-1.67,-0.12)	0.025	-0.86(-1.65,-0.07)	0.034	Per $10 \mu\text{g}/\text{m}^3$	-0.64(-1.22,-0.05)	0.033	-0.67(-1.28,-0.06)	0.032
Q1 (≤ 26.18)	Reference		Reference		Q1 (≤ 25.59)	Reference		Reference	
Q2 (26.18~)	-0.19(-1.30,0.91)	0.732	-0.40(-1.52,0.73)	0.490	Q2 (25.59~)	0.07(-0.83,0.98)	0.872	0.04(-0.90,0.97)	0.938
Q3 (29.86~)	-1.39(-2.51,-0.27)	0.016	-1.21(-2.37,-0.05)	0.042	Q3 (29.41~)	-0.71(-1.62,0.20)	0.125	-0.74(-1.75,0.27)	0.150
Q4 (≥ 32.74)	-1.35(-2.45,-0.26)	0.017	-1.29(-2.41,-0.17)	0.025	Q4 (≥ 32.68)	-0.65(-1.56,0.25)	0.159	-0.73(-1.67,0.22)	0.134

^a Adjusted for maternal age, parity, gestational weeks, paternal smoking, prepregnancy BMI, gestational hypertension, gestational diabetes, adverse pregnant history and infant sex.

^b The mean of weekly PM_{10} , $\text{PM}_{2.5}$ or PM_1 exposure during the susceptible time in the first nested case-control study.

^c The mean of weekly PM_{10} , $\text{PM}_{2.5}$ or PM_1 exposure during the susceptible time in the second nested case-control study.

^d Cases and controls were selected from the participants whose maternal blood samples have been collected.

^e Cases and controls were selected from the participants whose cord blood samples have been collected.

3.4. Associations between LINE-1 methylation and PTB in the nested case-control studies

Table 5 shows the associations between maternal and cord blood LINE-1 methylation with risk of PTB. We observed inverse association patterns of maternal blood and cord blood LINE-1 methylation with PTB, but the associations were not statistically significant. The risk of PTB was negatively associated with maternal blood LINE-1 methylation level (OR = 0.97 for per 1%5mC, 95%CI: 0.87–1.09), but positively correlated with cord blood LINE-1 methylation level (OR = 1.06 for per 1%5mC, 95%CI: 0.95–1.17). Similar results were found in the analyses in which the LINE-1 methylation levels were divided by the median (Table 5).

3.5. Sensitivity analysis

We changed the *dfs* of lag distribution of PM exposure in the DLNM models, and found that our results did not significantly change (Fig. 2). The results of the two-pollutants models show that additional adjustment for O₃ did not substantially change our findings in the single-pollutant models (Fig. S3). However, significantly negative associations of PM_{2.5} and PM₁ with PTB risk were observed during the 25th to 28th gestational weeks after adjustment for SO₂ exposure. Moreover, the effects of PM_{2.5} and PM₁ exposure during the susceptible exposure windows were no longer significant, which may be due to the close correlations of PM_{2.5} and PM₁ with CO concentrations (Table S5). The effect patterns of PM on preterm were significantly changed after adjustment for the weekly ambient temperature, indicating the potential confounding effects of temperature in the associations between PM and PTB (Fig. S4). In addition, the performance of the spatiotemporal LUR modeling on air pollutants did not substantially change with the changes in the buffer radiuses. For example, for the R² of LUR modeling on PM_{2.5}, the assessment ranged from 87.47% to 88.92%, and the RMSE ranged between 5.12 µg/m³ and 5.88 µg/m³. All these results indicated the robustness of the spatiotemporal LUR modeling to the radiuses (Tables S6 to S11).

4. Discussion

In this study, we employed a birth cohort study to investigate the associations between particulate air pollution exposure during pregnancy and PTB, and further assessed the role of LINE-1 methylation. We observed a more pronounced increase in risk of PTB in participants with higher PM_{2.5} and PM₁ exposure, and the 12th to 20th gestational weeks might be the susceptible exposure window. In addition, decreased maternal blood LINE-1 methylation might be a pathway the particulate air pollution exerts. According to our review of the literature, this study was the first to assess the effects of ambient PM of different sizes on the risk of PTB and further illustrate the possible epigenetic mechanisms in

China. Our findings could help health care workers clearly understand the effects of prenatal exposure to fine PM on the risk of PTB, implement interventional measures to reduce maternal exposure to air pollution exposure during susceptible windows, and reduce the risk of PTB. In addition, the roles of epigenetic alternation in the effects of air pollution may help develop pharmaceutical interventions to mitigate the risk of PTB.

The effects of prenatal exposure to PM_{2.5} on PTB risk have been assessed in many studies conducted mainly in developed countries, and most have observed increased risk of PTB induced by higher exposure to PM_{2.5} (Li et al., 2017; Malley et al., 2017; Sun et al., 2015; Zhu et al., 2015; Sheridan et al., 2019), which is consistent with our findings. For instance, Kloog et al. observed that the OR for PTB was 1.06 (95% CI: 1.01, 1.13) for each 10 µg/m³ increase in PM_{2.5} exposure during the entire pregnancy period (Kloog, 2012). Several studies conducted in China have also found significant effects of PM_{2.5} on PTB risk (Fleischer et al., 2014; Liu et al., 2019a, 2019b; Li et al., 2018; Cheng et al., 2016; Guo et al., 2018; Wang et al., 2018a, 2018b). In Fleischer et al.'s multi-country study, they employed remote sensing data to assess PM_{2.5} exposure in Chinese pregnant women, and found a significantly higher risk (OR = 2.54, 95%CI: 1.42–4.55) of preterm for participants who had the highest quartile of PM_{2.5} exposure (> 36.5 vs < 12.5 µg/m³) (Fleischer et al., 2014). One of our previous studies applied the birth registry data in Guangzhou and found positive associations between PM_{2.5} exposure (HR = 1.29 for 27 µg/m³ (IQR), 95%CI: 1.02–1.64) during the second trimester with moderate PTB risk (Wang et al., 2018a, 2018b). However, most studies have been ecological studies, retrospective studies, or studies based on birth registry data. These studies have not been able to sufficiently adjust for confounding factors, which may lead to biased results. In this study, we employed a perspective cohort study design to collect the exposure and outcome information, and sufficiently adjusted for potential confounding factors. Therefore, our study provided more solid evidence on the effects of PM on PTB risk than has been presented in the literature.

Although we did not observe significant effects of PM₁₀ exposure, we observed a slightly greater effect of PM₁ than PM_{2.5} exposure on PTB risk. Few studies have investigated the effects of PM₁ on PTB risk due to the unavailability of data. A recent national level study in China using the birth registry data observed significant association between maternal PM₁ exposure during the entire pregnancy and increased risk of PTB (Wang et al., 2018a, 2018b). As we found in this study, they also found more pronounced effects of PM with smaller sizes (Li et al., 2018; Wang et al., 2018a, 2018b). PM₁ is the major component (approximately 80%) of the PM_{2.5} that contributes to nearly 65% of ambient PM₁₀ in China (Wang et al., 2015; Zhou et al., 2016). Smaller particles, especially PM₁, easier to inhale into the acinar part of the respiratory tract, penetrate into the gas exchange region, invade blood circulation, and activate multiple pathophysiological processes including oxidative stress, inflammation and DNA damage, which may trigger the

Table 5
Associations (OR, 95%CI) of LINE-1 methylation with PTB in the nested case-control studies.

LINE-1 methylation levels	Crude OR(95%CI)	Adjusted OR(95%CI) ^a	Adjusted OR(95%CI) ^b
Nested case-control study I ^c			
Per 1%5mC increase	1.01(0.91,1.13)	0.98(0.87,1.10)	0.97(0.87,1.09)
≤ P ₅₀ (84.72%5mC)	1	1	1
> P ₅₀ (84.72%5mC)	0.85(0.47,1.53)	0.78(0.42,1.49)	0.73(0.38,1.41)
Nested case-control study II ^d			
Per 1%5mC increase	1.06(0.96,1.15)	1.05(0.95,1.16)	1.06(0.95,1.17)
≤ P ₅₀ (85.04%5mC)	1	1	1
> P ₅₀ (85.04%5mC)	1.42(0.86,2.32)	1.52(0.89,2.60)	1.59(0.92,2.75)

^a Adjusted for maternal age, infant sex, parity, household income, prepregnancy BMI, passive smoking, gestational hypertension and tea consumption.

^b Adjusted for a and marital status, gravidity, maternal education and gestational diabetes.

^c Cases and controls were selected from the participants whose maternal blood samples have been collected.

^d Cases and controls were selected from the participants whose cord blood samples have been collected.

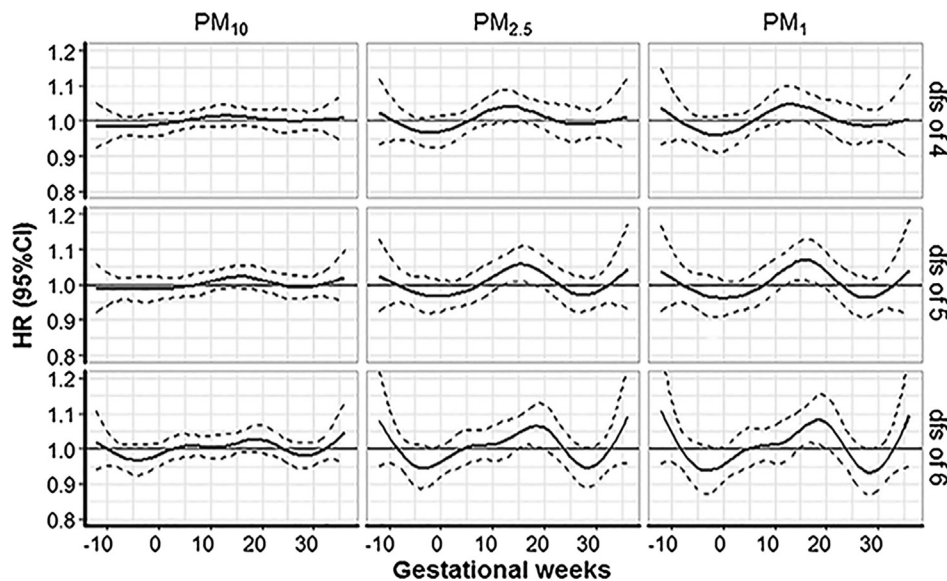


Fig. 2. Sensitivity analyses on the impacts of dfs of the lag structure in the DLNM models on the associations between weekly PM exposure (per 10 $\mu\text{g}/\text{m}^3$ increment) during pregnancy and PTB. HRs (95%CI) indicate the risks of PTB for each 10 $\mu\text{g}/\text{m}^3$ increment in weekly PM concentrations during pregnancy.

Zero in the X-axis scale indicates the time of LMP (gestational week 0), minus numbers indicate the weeks prior to LMP, and plus numbers indicate the gestational weeks.

All models were adjusted for maternal age, gravidity, parity, infant sex, maternal occupation, household income, maternal education, marital status, season of conception, prepregnancy BMI, maternal smoking, passive smoking, paternal smoking, gestational hypertension, gestational diabetes, adverse pregnancy history, alcohol consumption, tea consumption, and weekly mean temperature.

occurrence of PTB (Chen et al., 2017; Filep et al., 2016). In addition, smaller particles can remain in the air for a longer time increasing the likelihood of exposure (Srimuruganandam and Shiva Nagendra, 2012), and carry more deleterious substances, such as oxidant gaseous pollutants, organic compounds, and transition metals, due to the larger specific surface area (Brook et al., 2004). Therefore, more attention should be paid to finer particles. Our findings suggest that integrate ambient PM_{10} and other ultrafine particles should be integrated into the national air quality monitoring system in China.

Studies have examined the critical windows of maternal air pollution exposure using different strategies including assessment by trimesters (Schifano et al., 2016; Symanski et al., 2016; Zhang et al., 2012), months (Le et al., 2012) or specific gestational weeks (Chang et al., 2015; Warren et al., 2013). However, susceptible air pollution exposure windows remain controversial. Trimester-average exposure has been the most commonly used approach to assess the effects of air pollution exposure. However, the exposure in different trimesters has usually closely correlated with each other, and the effect of exposure during a given trimester without adjusting for exposure in other trimesters can result in biased estimates. Wilson et al. (Wilson et al., 2017) compared different strategies and suggested that critical exposure windows may not align with clinically defined trimesters, and that the DLM model can eliminate the bias.

In this study, we employed a DLNM model in which the weekly PM exposure was introduced to identify susceptible exposure windows. We observed that the 12th to 20th gestational weeks might be the critical window, with a peak effect in the 16th gestational week. This period covers the late period of the first trimester and a substantial part of the second trimester, which is consistent with two studies that have assessed the weekly-specific effects of air pollution (Chang et al., 2015; Warren et al., 2013). Animal and human studies have shown that, in order to fulfill the increasing demand for oxygen and nutrients, the umbilical artery blood flow rate and blood flow increase starting in from the second trimester of pregnancy, which also increases fetal exposure to other exogenous factors such as air pollution (Ha et al., 2014; Blum et al., 2017; Wang et al., 2018a, 2018b). Therefore, disruption of air pollution during the 12th to 20th gestational weeks might be more likely to lead to PTB. These results can assist with defining underlying mechanisms and guiding prenatal care. Clinical workers could focus on health education for pregnant women during critical periods to reduce exposure to air pollutants and, thereby, reduce the risk of PTB.

We observed that maternal LINE-1 methylation was negatively associated with maternal exposure to PM. Maternal LINE-1 methylation

was also negatively correlated with the risk of PTB but was not statistically significant. These results indicate that LINE-1 methylation alteration might be an underlying mechanism of PM increasing PTB risk. Our findings were consistent with several studies that have observed negative associations between PM levels and LINE-1 methylation (Alfano et al., 2018). For example, Bram et al. found that placental global DNA methylation was inversely associated with $\text{PM}_{2.5}$ exposure during the whole pregnancy (Janssen et al., 2013). Cai et al. (2017) found early pregnancy PM_{10} exposure was reversely associated with placental LINE-1 methylation. PM increased the production of reactive oxygen species (Donaldson et al., 2001) which may disrupt DNA and, thus, interfere with methylation (Valinluck et al., 2004). Additionally, LINE-1 hypomethylation can induce the increase of activation and transcription of retrotransposons, which may trigger the immune system to induce inflammatory reaction (Crow, 2010), and initiate the mechanisms for PTB (Felipe et al., 2014). LINE-1 hypomethylation can also affect cellular function by enhancing the transcription of sequences initiated during conditions of inflammation (Baccarelli et al., 2010).

Positive associations between cord blood LINE-1 methylation and PTB risk were found in this study, which does not support our hypothesis. A prospective cohort study also observed that PTB was associated with higher LINE-1 methylation in cord blood (Burris et al., 2012). The reasons for this difference remain unknown. Our findings may indicate that the effects of maternal and fetal DNA methylation on the PTB risk may differ from one another, and the regulatory mechanisms of LINE-1 methylation in cord blood during fetal development may differ from maternal LINE-1 methylation during adulthood. A study conducted by Ladd-Acosta et al. (Ladd-Acosta et al., 2019) also found different DNA methylation levels in cord blood and placenta associated with prenatal exposure to air pollutants. Therefore, further research would further illustrate these differences.

4.1. Strengths and limitations

One advantage of our study is that we conducted a prospective birth cohort study, which can provide a stronger causal argument on the effects of prenatal PM exposure on the risk of PTB. The detailed individual information of participants from multiple sources including a questionnaire interview, medical records, and other environmental factors allowed us to sufficiently adjust for the potential confounding effects. Second, this study simultaneously estimated the effects of PM of different sizes, which can directly assess the impacts of different particle sizes on the effect estimations. Third, we employed a

spatiotemporal LUR model incorporating each individual's activity patterns and the air pollutants' infiltration factors to assess each individual level of exposure to air pollutants, which could substantially decrease the misclassification bias. The performance of the spatiotemporal LUR model was better than in some previous studies (Hoogh et al., 2016; Mao et al., 2012). For example, Hoogh et al. employed a LUR model incorporating satellite-derived and chemical transport modeling data to estimate PM_{2.5} in Western-Europe, and found that the R² was 58%. Fourth, we employed a DLNM model incorporating a Cox PH model to estimate the weekly-specific prenatal exposure to PM on PTB risk. This advanced method eliminates the confounding bias and provides a more reliable susceptible exposure window. Finally, we simultaneously tested LINE-1 methylation in maternal and cord blood DNA, which allowed us to compare their different associations with PTB.

Several limitations of our study should be considered. First, only one hospital was used as the study setting, which may limit the generalization of our findings. In further research, we plan to conduct a multicenter study to further attest these findings. Second, although the spatiotemporal LUR models could predict PM₁₀ and PM_{2.5} concentrations well, the performance of predicting gaseous pollutant such as SO₂, NO₂, and CO, indicated that more predictable variables are necessary in the LUR models. A recent study found that a comprehensive Kruskal-K-means method that clustered the measurement samples from multiple heterogeneous sources could substantially improve the prediction of NO₂, and the R² was 88% (Li et al., 2019). Therefore, more research is necessary to improve the prediction of gaseous pollutants. Third, the number of PTB cases used to measure LINE-1 methylation was relatively small. Hence the statistical powers in the nested case-control analyses were low, which may be a reason for the absence of a significant association between LINE-1 methylation and PTB. Additionally, the relatively small sample prohibited us from testing the associations between PM exposure and LINE-1 methylation by using a DLNM model. Finally, we tested the global methylation level indicated by LINE-1 methylation. Gene-specific or genome-wide DNA methylation may provide more information on the mechanisms underlying the effects of air pollution on PTB risk. Therefore, further research could test the epigenetic mechanisms.

5. Conclusions

We found that higher maternal prenatal exposure to PM₁ and PM_{2.5} was associated with increased risk of PTB, and PM with smaller sizes may have more pronounced effects. The 12th to 20th gestational weeks might be the susceptible exposure window of PM₁ and PM_{2.5}. Maternal and fetal DNA LINE-1 methylation might be an underlying mechanism of PM increasing PTB risk. Further studies are required to understand the mechanisms.

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Declaration of competing interest

The authors declare no conflict of actual or potential competing financial interests.

Data available statement

Data are available on application to the Chinese Clinical Trial Registry from June 2021 (<http://www.chictr.org.cn/index.aspx>).

Ethics approval

The PEOH study was approved by the Ethics Committee of Guangdong Provincial Center for Disease Control and Prevention (W96-027E-2015001). All participants gave informed consent before taking part.

Contributors

XL (Xin Liu) and YY (Yufeng Ye) are joint first authors. TL (gzzt_2002@163.com) and BZ (cnzhangbo@126.com) designed this work and contributed equally to the correspondence work. XL (Xin Liu) and YY (Yufeng Ye) analyzed the data and drafted the manuscript. XL (Xiaona Li), WM, ZY, YG and YY (Yi Yang) edited the manuscript. XL (Xin Liu), XL (Xiaona Li), YC, BF, GC, JX, WZ, XL (Xing Li), JS, DN, QW, YZ, and QD contributed to data collection and preparation. All authors contributed to further drafts and approved the final manuscript. TL and BZ are study guarantors. The corresponding author attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted.

Patient and public involvement

No patients were involved in setting the research question or outcome measures, nor were they involved in developing plans for design or implementation of the study.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envint.2019.105177>.

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